



Still on the Walk Sheet: A Controlled Interrupted Time-Series Study of Orphaned Quarterly Meter Reads After Automated Address Normalisation

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Abstract. Customer-record deduplication is ordinarily evaluated against the record set it cleans. We evaluate it instead against the physical asset the record stands for. A metropolitan water retailer enabled automated address normalisation across 218,400 metered residential connections, merging 1,412 account pairs whose normalised address strings collided. Meter readers continued to visit every merged property, because the quarterly walk sheet is generated from the meter asset register rather than the customer register, and neither the merge nor its reversal writes a row that the walk sheet can read. We report a controlled interrupted time-series analysis over sixteen quarters, with a comparison series drawn from 26,380 connections whose address strings fell below the parser's confidence threshold and were excluded from the merge pass. The rate of reads logged against no active account rises by 5.02 per 1,000 connections per quarter at the release (95% CI 4.41 to 5.63), with a slope change of 0.14 (95% CI 0.02 to 0.26); the comparison series shows no level change (0.08, 95% CI -0.21 to 0.37). The estimate is insensitive to every specification we considered. A field audit of 96 orphaned properties found 91 meters in service and registering flow. Over the same period the retailer's Record Integrity Index rose from 0.941 to 0.997 and has since held, and 71.4% of the customer contacts arising from the merges were closed under a code that removes the property from the indicator's own candidate queue.

Introduction

A water meter sits in a box in the front yard, under a hinged lid, a few metres in from the footpath. Once a quarter someone lifts the lid, reads five digits, and moves on to the next property. The read travels to a billing system, becomes a line on a bill, and is thereafter the only account anyone keeps of how much water that household used.

Between the meter and the bill sits a customer record, and customer records accumulate duplicates. Deduplication is a mature discipline with a settled theoretical basis [1], a large literature on making it scale [2], [3], and a well-understood place in data-quality practice [4], [5]. Address strings are among the hardest fields it operates on: the same dwelling is written a dozen ways,

and normalising those variants into a comparable form is a standing problem in geocoding as well as in linkage [6].

What the deduplication literature evaluates, almost without exception, is the record set. A merge is scored by whether the two records referred to the same entity. The question we ask here is narrower and, we think, more awkward: when the merge is wrong, what happens to the physical object the record was standing for, and how long does it take anyone to notice. Our contributions are, we suggest, threefold:

- a controlled interrupted time-series estimate of orphaned meter reads following an automated normalisation release, using a comparison cohort the release could not reach, which may offer a cleaner counterfactual than the

before-and-after audits the sector currently relies on;

- a field audit establishing that the orphaned meters remained in service and continued to be read, which we believe supports the stronger claim that the loss is one of attribution rather than of measurement;
- an account of how the retailer's own record-integrity indicator responded, which appears to generalise beyond this release.

Background

Probabilistic linkage assigns each candidate pair a weight and a decision under a pair of thresholds [1], and the practical literature is largely concerned with the consequences of setting those thresholds badly [7]. In administrative settings the errors are known to be consequential and hard to characterise after the fact [8]. Blocking and indexing schemes determine which pairs are compared at all [2], and a normalisation step that collapses surface variation in an address field necessarily raises the number of pairs that reach comparison in the first place.

Whether a merge can later be examined is a provenance question. The database literature distinguishes why a record is present from where its values came from [9], and surveys the machinery required to answer either [10]. A merge implemented as an in-place update to a surviving row answers neither: the merged-away identifier is not retained, so the operation is not addressable by any query the system supports.

The metering literature has moved decisively toward continuous telemetry, where consumption is observed often enough that anomalies surface within days [11], [12]. Quarterly manual reading remains widespread, and its failure mode is different in kind: a quarterly series cannot distinguish a property that stopped consuming from a property whose reads stopped arriving anywhere.

Finally, an indicator constructed from the record set is subject to the reactivity that indicators generally exhibit: measures reshape the activity they record [13], and targets attached to them reliably produce behaviour that satisfies the measure rather than the intent [14]. The categories a system enforces are rarely neutral about what can subsequently be said [15], and

transparency about a number is not the same as accountability for what it stands for [16].

Prior work at this institution bears on both halves. An audit of a volunteer bike-marshal fault checklist found it silently duplicating every filed report into two further queues, inflating whichever backlog dashboard was consulted [17]. An audit of a residence-hall laundry application found its displayed credit balance drifting from independently metered wash cycles across a payment-system migration [18]. Both describe a record and its referent parting company at a system boundary, and neither found the parting visible from inside the record.

Method

Setting. One metropolitan water retailer, 218,400 metered residential connections across eight supply zones, quarterly manual reading by a contracted field workforce. Walk sheets are generated from the meter asset register, which holds one row per physical meter and is keyed by property identifier; billing is driven by the customer register, keyed by account number. The two registers are joined nightly.

The release. At the end of the eighth observation quarter the retailer enabled an automated address-normalisation routine in the customer register. The routine applies five transformations in fixed order — case folding, whitespace collapse, ordinal-suffix normalisation, street-type expansion, and unit-separator stripping — and merges any account pair whose transformed address strings become identical and whose surname tokens overlap. The merge writes the union of the billing history to the surviving account and returns the merged-away account number to the allocation pool. No provenance row is retained.

Cohorts. The exposed cohort is the 192,020 connections whose address strings the parser accepted. The comparison cohort is the 26,380 connections whose strings fell below the parser's confidence threshold — non-standard rural addresses, multi-tenancy sites, and properties recorded against a lot number — and which were therefore excluded from the merge pass. Both cohorts share the reading contractor, the walk-sheet generator, the billing cycle, and the nightly join.

Outcome. An *orphaned read* is a read logged against a meter in the asset register that re-

solves to no active account at the nightly join. The primary outcome is orphaned reads per 1,000 connections per quarter, computed separately per cohort over sixteen quarters, eight before the release and eight after.

Analysis. Segmented regression on the quarterly series [19], [20], of the form

$$Y_t = \beta_0 + \beta_1 t + \beta_2 D_t + \beta_3 (t - t_0) D_t + \gamma C_t + \varepsilon_t$$

where D_t indicates the post-release period, t_0 the release quarter, and C_t the contemporaneous comparison-cohort rate. β_2 is the level change and β_3 the slope change. Newey–West standard errors accommodate residual autocorrelation, and a pair of seasonal harmonics absorbs the reading calendar. The controlled specification follows the standard treatment [21]; the uncontrolled series was analysed first, per the tutorial procedure [22]. The release quarter itself was excluded as a transition period and restored in a sensitivity analysis.

Field audit. A random sample of 96 properties was drawn from the orphaned set at quarter sixteen and visited over three weeks by two auditors working independently of the retailer. Auditors recorded whether a meter was present, whether the register was advancing under a controlled two-minute draw, and whether the most recent logged read for that property appeared against any account.

Results

Cohort characteristics are given in Table 1.

Cohort	Connections	Unit-numbered	Merges
Exposed	192,020	31.4%	1,412
Comparison	26,380	9.8%	0

Table 1: Cohort composition at the release quarter. The comparison cohort is smaller and less unit-dense than the exposed cohort; no connection in it was merged during the observation window.

The release is visible in every supply zone at once (Figure 1). Before it, orphaned reads occur at a low and stable rate attributable to manual merges; after it, the rate rises in proportion to each zone’s share of unit-numbered addresses.

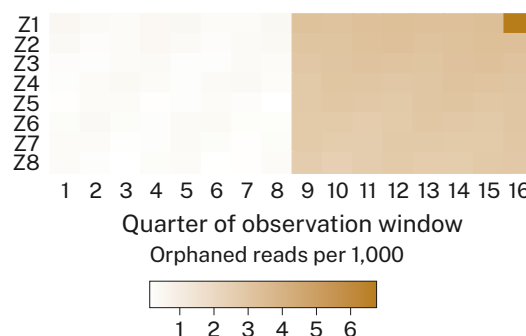


Figure 1: Orphaned reads per 1,000 connections, by supply zone and quarter. Every zone steps up between quarters 8 and 9; no zone steps up at any other quarter, and no zone returns to its pre-release rate within the window.

The pre-specified level change is 5.02 orphaned reads per 1,000 connections per quarter (95% CI 4.41 to 5.63), with a slope change of 0.14 per quarter (95% CI 0.02 to 0.26). The comparison cohort shows a level change of 0.08 (95% CI -0.21 to 0.37) and no slope change. The pre-release trend is flat in both cohorts.

Of the 1,412 merges executed in the release quarter, the largest group arose from the last transformation applied (Figure 2). Unit-separator stripping collapses the distinction between a sub-address and a street number written with a space, which is the collision the exposed cohort’s unit density predicts.

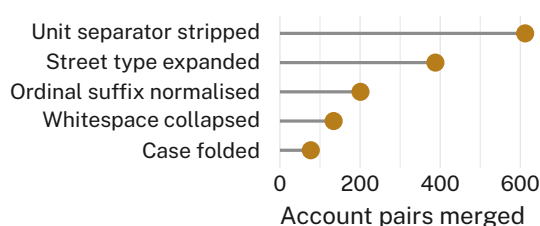


Figure 2: Account pairs merged in the release quarter, by the transformation whose output produced the collision. Unit-separator stripping accounts for 43.3% of merges on its own, and the two most permissive transformations together account for 70.8%.

The field audit visited 96 properties from the orphaned set. Meters were present at 94, in service and registering flow under a controlled draw at 91, and carried a logged read from the most recent quarter at 89. None of the 89 reads appeared against any account. The two properties without a meter had been demolished, and both remained on the walk sheet.

At quarter sixteen, 1,147 meters remained orphaned. Their disposition is shown in Figure 3. The largest group is the one requiring no action from anyone: read on schedule, logged on schedule, posted nowhere.

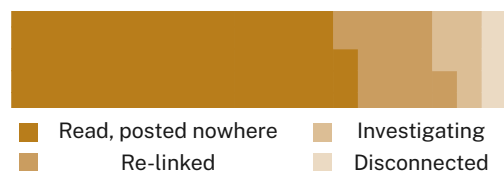


Figure 3: Disposition of the 1,147 orphaned meters at quarter sixteen. Two thirds are still being read to no destination; every re-linked meter in the second group was re-linked following a customer contact, and none was identified by the retailer first.

Customer contact was neither prompt nor universal. Among orphaned properties that contacted the retailer, the median interval from merge to first contact was two quarters — that is, after a second bill failed to arrive — and 38.1% made no contact at all within the eight post-release quarters. Of the contacts received, 71.4% were closed under the code *duplicate resolved*, which removes the property from the candidate-duplicate queue from which the Record Integrity Index is computed. The index rose from 0.941 in the release quarter to 0.997 two quarters later and has since held.

Five specifications were compared over both post-release windows (Figure 4). Dropping the seasonal term moves the level change by 0.07 per 1,000 (n.s.); dropping the autocorrelation correction moves it by 0.06 in the other direction (n.s.); dropping the comparison series entirely moves it by 0.09 (n.s.); retaining the transition quarter moves it by 0.08 (n.s.). We retain all four terms for completeness.

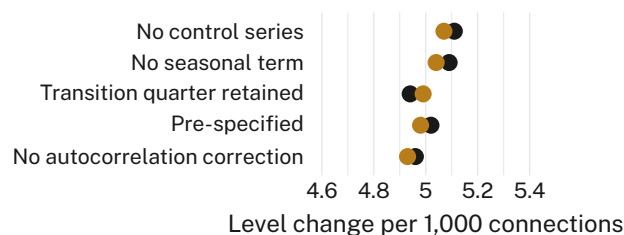


Figure 4: Level-change estimate under five specifications and two post-release windows. No pair among the ten estimates differs by more than 0.18 per 1,000, and the specification without a comparison series returns the largest estimate of the five.

Discussion

We observe a consistent level shift across zones, windows, and specifications. The normalisation release appears to have separated a population of meters from the accounts that billed them, without interrupting either the reading of those meters or the logging of the reads. The water was delivered, the meter turned, the reader attended, and the number was written down. Only the last step in the chain — the one that attaches a number to a household — was absent, and it is the only step no field process is designed to check.

The indicator's behaviour follows from its construction rather than from any decision about it. The Record Integrity Index counts candidate duplicate pairs against active accounts, so a merge reduces the numerator whether or not the two records described the same dwelling, and the closure code applied to a complaint about a wrong merge reduces it again. The one population positioned to falsify the index is the population the index has already removed from view. That the merge writes no provenance row is what makes this stable rather than self-correcting: an operation that cannot be queried cannot be audited [10], and a number computed over what remains will report improvement either way.

Limitations

The study covers a single retailer and a single normalisation release, though the consistency of the estimate across all eight supply zones and both post-release windows suggests the result may travel to comparable utilities. Our comparison cohort differs systematically from the exposed cohort in address form, which is the very property that determined exposure;

the specification without a comparison series returns a slightly larger estimate, so the control if anything understates the effect. The field audit was drawn at a single time point and cannot speak to meters orphaned and re-linked earlier in the window, which we would expect to be the better-handled cases.

Conclusion

A controlled interrupted time-series analysis across sixteen quarters finds that an automated address-normalisation release orphaned meters at a rate of roughly five per 1,000 connections per quarter, that the affected meters continued to be read on schedule, and that the retailer's record-integrity indicator improved throughout. The reads are still being taken; the walk sheet has never been told otherwise. Future work might follow a single orphaned meter through a full renewal cycle, and establish whether an asset that appears on no account can be retired from the register that still schedules its reading.

References

- [1] I. P. Fellegi and A. B. Sunter, "A Theory for Record Linkage," *Journal of the American Statistical Association*, vol. 64, no. 328, pp. 1183–1210, 1969, doi: [10.1080/01621459.1969.10501049](https://doi.org/10.1080/01621459.1969.10501049).
- [2] P. Christen, "A Survey of Indexing Techniques for Scalable Record Linkage and Deduplication," *IEEE Transactions on Knowledge and Data Engineering*, vol. 24, no. 9, pp. 1537–1555, 2012, doi: [10.1109/TKDE.2011.127](https://doi.org/10.1109/TKDE.2011.127).
- [3] A. K. Elmagarmid, P. G. Ipeirotis, and V. S. Verykios, "Duplicate Record Detection: A Survey," *IEEE Transactions on Knowledge and Data Engineering*, vol. 19, no. 1, pp. 1–16, 2007, doi: [10.1109/TKDE.2007.250581](https://doi.org/10.1109/TKDE.2007.250581).
- [4] C. Batini, C. Cappiello, C. Francalanci, and A. Maurino, "Methodologies for data quality assessment and improvement," *ACM Computing Surveys*, vol. 41, no. 3, pp. 1–52, 2009, doi: [10.1145/1541880.1541883](https://doi.org/10.1145/1541880.1541883).
- [5] T. C. Redman, "The impact of poor data quality on the typical enterprise," *Communications of the ACM*, vol. 41, no. 2, pp. 79–82, 1998, doi: [10.1145/269012.269025](https://doi.org/10.1145/269012.269025).
- [6] P. A. Zandbergen, "A comparison of address point, parcel and street geocoding techniques," *Computers, Environment and Urban Systems*, vol. 32, no. 3, pp. 214–232, 2008, doi: [10.1016/j.compenvurbsys.2007.11.006](https://doi.org/10.1016/j.compenvurbsys.2007.11.006).
- [7] J. C. Doidge and K. Harron, "Demystifying probabilistic linkage," *International Journal of Population Data Science*, vol. 3, no. 1, 2018, doi: [10.23889/ijpds.v3i1.410](https://doi.org/10.23889/ijpds.v3i1.410).
- [8] K. Harron et al., "Challenges in administrative data linkage for research," *Big Data & Society*, vol. 4, no. 2, 2017, doi: [10.1177/2053951717745678](https://doi.org/10.1177/2053951717745678).
- [9] P. Buneman, S. Khanna, and W.-C. Tan, "Why and Where: A Characterization of Data Provenance," *Lecture Notes in Computer Science*, pp. 316–330, 2001, doi: [10.1007/3-540-44503-X_20](https://doi.org/10.1007/3-540-44503-X_20).
- [10] J. Cheney, L. Chiticariu, and W.-C. Tan, "Provenance in Databases: Why, How, and Where," *Foundations and Trends in Databases*, vol. 1, no. 4, pp. 379–474, 2009, doi: [10.1561/19000000006](https://doi.org/10.1561/19000000006).
- [11] A. Cominola, M. Giuliani, D. Piga, A. Castelletti, and A. E. Rizzoli, "Benefits and challenges of using smart meters for advancing residential water demand modeling and management: A review," *Environmental Modelling & Software*, vol. 72, pp. 198–214, 2015, doi: [10.1016/j.envsoft.2015.07.012](https://doi.org/10.1016/j.envsoft.2015.07.012).
- [12] T. C. Britton, R. A. Stewart, and K. R. O'Halloran, "Smart metering: enabler for rapid and effective post meter leakage identification and water loss management," *Journal of Cleaner Production*, vol. 54, pp. 166–176, 2013, doi: [10.1016/j.jclepro.2013.05.018](https://doi.org/10.1016/j.jclepro.2013.05.018).
- [13] W. N. Espeland and M. Sauder, "Rankings and Reactivity: How Public Measures Recreate Social Worlds," *American Journal of Sociology*, vol. 113, no. 1, pp. 1–40, 2007, doi: [10.1086/517897](https://doi.org/10.1086/517897).
- [14] G. Bevan and C. Hood, "What's Measured Is What Matters: Targets and Gaming in the English Public Health Care System," *Public Administration*, vol. 84, no. 3, pp. 517–538, 2006, doi: [10.1111/j.1467-9299.2006.00600.x](https://doi.org/10.1111/j.1467-9299.2006.00600.x).

- [15] G. C. Bowker and S. L. Star, *Sorting Things Out: Classification and Its Consequences*. The MIT Press, 1999. doi: [10.7551/mitpress/6352.001.0001](https://doi.org/10.7551/mitpress/6352.001.0001).
- [16] M. Strathern, “The Tyranny of Transparency,” *British Educational Research Journal*, vol. 26, no. 3, pp. 309–321, 2000, doi: [10.1080/713651562](https://doi.org/10.1080/713651562).
- [17] J. Nwosu and T. Solberg, “Filed Once, Counted Thrice: Auditing a Volunteer Bike-Marshall Fault Checklist’s Silent Fan-Out Across Three Facilities Queues,” *Slop University technical report*, 2026, doi: [10.5555/slop.a1axjp](https://doi.org/10.5555/slop.a1axjp).
- [18] C. Beng, M. Hanke, and R. Sabel, “Displayed, Not Metered: Auditing a Residence-Hall Laundry App’s Credit Balance Against Machine-Metered Wash Cycles Across a Payment-System Migration,” *Slop University technical report*, 2026, doi: [10.5555/slop.7xifby](https://doi.org/10.5555/slop.7xifby).
- [19] A. K. Wagner, S. B. Soumerai, F. Zhang, and D. Ross-Degnan, “Segmented regression analysis of interrupted time series studies in medication use research,” *Journal of Clinical Pharmacy and Therapeutics*, vol. 27, no. 4, pp. 299–309, 2002, doi: [10.1046/j.1365-2710.2002.00430.x](https://doi.org/10.1046/j.1365-2710.2002.00430.x).
- [20] E. Kontopantelis, T. Doran, D. A. Springate, I. Buchan, and D. Reeves, “Regression based quasi-experimental approach when randomisation is not an option: interrupted time series analysis,” *BMJ*, vol. 350, p. h2750, 2015, doi: [10.1136/bmj.h2750](https://doi.org/10.1136/bmj.h2750).
- [21] J. Lopez Bernal, S. Cummins, and A. Gasparrini, “The use of controls in interrupted time series studies of public health interventions,” *International Journal of Epidemiology*, vol. 47, no. 6, pp. 2082–2093, 2018, doi: [10.1093/ije/dyy135](https://doi.org/10.1093/ije/dyy135).
- [22] J. Lopez Bernal, S. Cummins, and A. Gasparrini, “Interrupted time series regression for the evaluation of public health interventions: a tutorial,” *International Journal of Epidemiology*, vol. 46, no. 1, pp. 348–355, 2017, doi: [10.1093/ije/dyw098](https://doi.org/10.1093/ije/dyw098).